HERO HOUSING FINTECH CHALLENGE · PROBLEM STATEMENT 2 · I-HACK · IIT BOMBAY E-SUMMIT 2026

गृह Griha

Mortgage underwriting you can explain, replay and defend to a regulator.

Home Loan and LAP decisions in about 10 ms, with reasons, an RBI-format Key Facts Statement and an audit trail.

Live d3l1fg79dbflis.cloudfront.net

Code github.com/anshul23102/griha

TeamBreezy

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THE PROBLEM AND OUR ANSWER

Home loans are slow, opaque and heavily regulated. Griha decides the clear cases alone and shows its work on every one.

10 ms
per decision, median engine time

62.5%
decided with no person involved (1,000 synthetic cases)

19
policy rules as code, hashed into every decision

31,347
sanctions-list entries screened on every case

Explainable
Reasons, evidence and a Path to Yes

Regulator-aware
Each control cites its paragraph

Honest
Real, simulated and not built, labelled

Language models only read documents. They never decide: every outcome comes from a hashed rulebook and a calibrated, monotone risk model.

END TO END

One application, eight stations, one decision.

1	2	3	4	5	6	7	8
Data	Income	Property	Eligibility	Policy	Risk	Fraud + KYC	Compliance

IN
Four-step form, PDF statement or Account Aggregator data. Consent first.

→

ABOUT 10 MS
Outcome, offer, reasons, Key Facts Statement, audit entry.

→

OUT
APPROVE with offer · **REJECT** with reasons · **REFER** to the right queue, evidence ready

A Temporal workflow runs the stations, retries a failed source and waits on referrals with an SLA timer. If a source stays down the case goes to a person; nothing is guessed.

REQUIREMENTS 1 AND 5 · DATA, DOCUMENTS, PROPERTY

Four sources feed the engine. Documents are verified, never obeyed.

Bank statement

REAL PARSER

PDF, or the ReBIT Account Aggregator file

Credit bureau

SIMULATED

Shaped like a bureau report

KYC + sanctions

LISTS REAL

UN, OFAC, EU, UK lists; V-CIP checklist

Land records

SIMULATED

Title, RERA, encumbrance

Signed

Account Aggregator, end to end: signed request, encrypted data, decrypted and read (sandbox)

6.3%

Property valuation median error (synthetic); loans from ₹50 lakh need two reports

0

Instructions obeyed from inside a document: text aimed at an AI reviewer is flagged as tampering

REQUIREMENTS 2 AND 4 · POLICY AND RISK

Policy is code. Risk is a model that cannot contradict common sense.

12 inputs
FOIR, LTV, bureau, income variability...

→

Monotone LightGBM
worse never lowers risk

→

Calibrated
3% means about 3%

→

SHAP reasons
and a 650-point scorecard

0.788

AUC on **real** German borrowers, with constraints (0.793 without)

0.782

AUC on **real** Taiwan borrowers, 30,000 (0.783 without)

+5pt

auto-approvals from loosening FOIR 55 to 65%, but defaults rise 3.4 to 4.1%: the Policy Simulator shows both

The boosted model does not beat plain logistic regression on our synthetic data (0.747 vs 0.760); we say so and keep it for the monotone guarantee. UCI datasets, CC BY 4.0: not Indian mortgages.

REQUIREMENT 3 · FRAUD AND KYC / AML

Every case is screened, and the law decides what the applicant sees.

Detect

- Edited statements
- Round-trip money
- Linked-applicant rings
- Behaviour anomalies

119 of 120 injected fraud cases held back

Screen

31,347

entries on the real UN, US OFAC, EU and UK lists. A match goes to an officer, never an automatic decline

Withhold

Fraud and sanctions findings are hidden from the applicant (KYC Directions paras 19, 20(4), 52). Officers see everything.

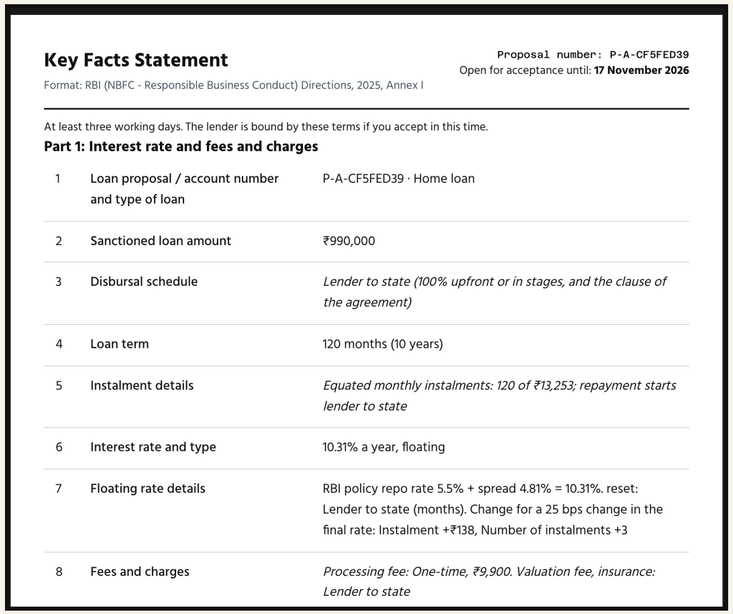
NEW

Video KYC (V-CIP) checklist: a session cannot be completed unless every mandatory element of paras 26 and 27 is on record. Prompting forces rejection.

REQUIREMENT 6 · COMPLIANCE

Five regulatory texts, read in full, drive the controls.

RBI HFC DIRECTIONS 2025 | DIGITAL LENDING 2025 | NBFC KYC 2025 | RESPONSIBLE BUSINESS CONDUCT 2025 | DPDP RULES 2025



Key Facts Statement

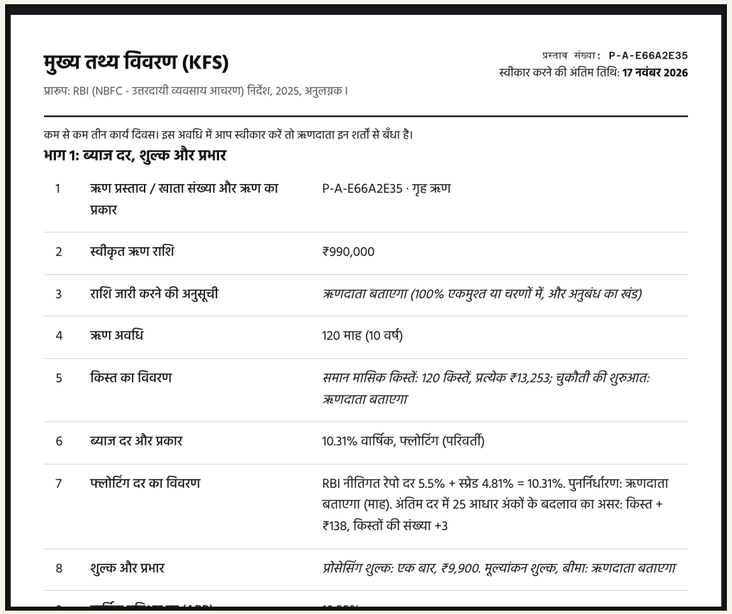
Format: RBI (NBFC - Responsible Business Conduct) Directions, 2025, Annex I

Proposal number: P-A-CF5FED39
Open for acceptance until: **17 November 2026**

At least three working days. The lender is bound by these terms if you accept in this time.

Part 1: Interest rate and fees and charges

1	Loan proposal / account number and type of loan	P-A-CF5FED39 · Home loan
2	Sanctioned loan amount	₹990,000
3	Disbursal schedule	*Lender to state (100% upfront or in stages, and the clause of the agreement)*
4	Loan term	120 months (10 years)
5	Instalment details	*Equated monthly instalments: 120 of ₹13,253; repayment starts lender to state*
6	Interest rate and type	10.31% a year, floating
7	Floating rate details	RBI policy repo rate 5.5% + spread 4.81% = 10.31%. reset: Lender to state (months). Change for a 25 bps change in the final rate: Instalment +₹138, Number of instalments +3
8	Fees and charges	*Processing fee: One-time, ₹9,900. Valuation fee, insurance: Lender to state*



मुख्य तथ्य विवरण (KFS)

प्रारूप: RBI (NBFC - उत्तरदायी व्यवसाय आचरण) निर्देश, 2025, अनुलग्नक I

प्रस्ताव संख्या: P-A-E66A2E35
स्वीकार करने की अंतिम तिथि: **17 नवंबर 2026**

कम से कम तीन कार्य दिवस। इस अवधि में आप स्वीकार करें तो ऋणदाता इन शर्तों से बँधा है।

भाग 1: ब्याज दर, शुल्क और प्रभार

1	ऋण प्रस्ताव / खाता संख्या और ऋण का प्रकार	P-A-E66A2E35 · गृह ऋण
2	स्वीकृत ऋण राशि	₹990,000
3	राशि जारी करने की अनुसूची	*ऋणदाता बताएगा (100% एकमुश्त या चरणों में, और अनुबंध का खंड)*
4	ऋण अवधि	120 माह (10 वर्ष)
5	किस्त का विवरण	*समान मासिक किस्तें: 120 किस्तें, प्रत्येक ₹13,253; चुकौती की शुरुआत: ऋणदाता बताएगा*
6	ब्याज दर और प्रकार	10.31% वार्षिक, फ्लोटिंग (परिवर्ती)
7	फ्लोटिंग दर का विवरण	RBI नीतिगत रेपो दर 5.5% + स्प्रेड 4.81% = 10.31%. पुनर्निर्धारण: ऋणदाता बताएगा (माह). अंतिम दर में 25 आधार अंकों के बदलाव का असर: किस्त + ₹138, किस्तों की संख्या +3
8	शुल्क और प्रभार	*प्रोसेसिंग शुल्क: एक बार, ₹9,900. मूल्यांकन शुल्क, बीमा: ऋणदाता बताएगा*

17.07%

Our APR reproduces the RBI's own worked example. A permanent test.

Key Facts Statement in the Annex I layout, English and Hindi. What only a lender knows reads "Lender to state". We never invent it.

Loan servicing: quarterly statement, document-release clock with compensation, recovery-contact rules.

REQUIREMENT 7 · INSTANT DECISIONING

Clear cases are decided alone. People see only what needs them.

APPROVE 53.8%	8.7%	REFER 37.5%

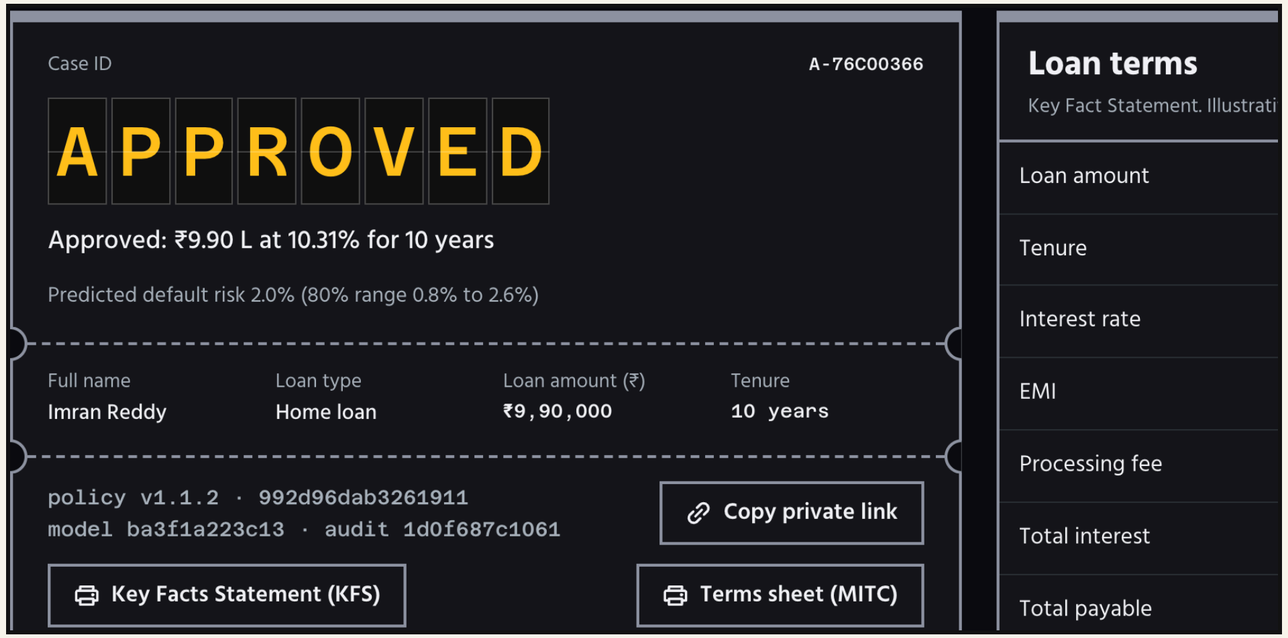
25.5%
Credit queue

5.1%
Fraud + compliance

4.9%
Property

2.0%
Data outage

We tried loosening the limits: automatic rate rose 62.5 to 69.8%, defaults of approved loans 3.4 to 4.9%. We left them; a lender sets them from its own losses.



REQUIREMENT 8 · EXPLAINABILITY AND GOVERNANCE

Every decision can be explained, traced and replayed.

1

Explain

Reasons with evidence, SHAP drivers, a **Path to Yes**, a letter in English or Hindi

2

Trace

Hash-chained audit log; replay any decision byte for byte; tamper detection shown live

3

Govern

Four-eyes approval, SLA timers, model card, drift and fairness checks, erasure by crypto-shredding

Real-data check

The same modelling recipe on two public datasets of real borrowers (UCI, CC BY 4.0). AUC, 5-fold out-of-sample.

Dataset	n	Logistic regression	Boosting, free	Boosting, monotone (ours)
German Credit (UCI, CC BY 4.0)	1,000	0.782	0.793	0.788
Taiwan credit-card default (UCI, CC BY 4.0)	30,000	0.723	0.783	0.782

Real outcomes, but not Indian mortgages: German instalment loans and Taiwanese cards. What it shows: the monotone constraints cost almost no ranking power, and the isotonic probabilities match observed default rates out of sample. What it does not show: accuracy on Indian home loans.

Governance: real-data check, with audit chain and replay on the same page

Portfolio impact

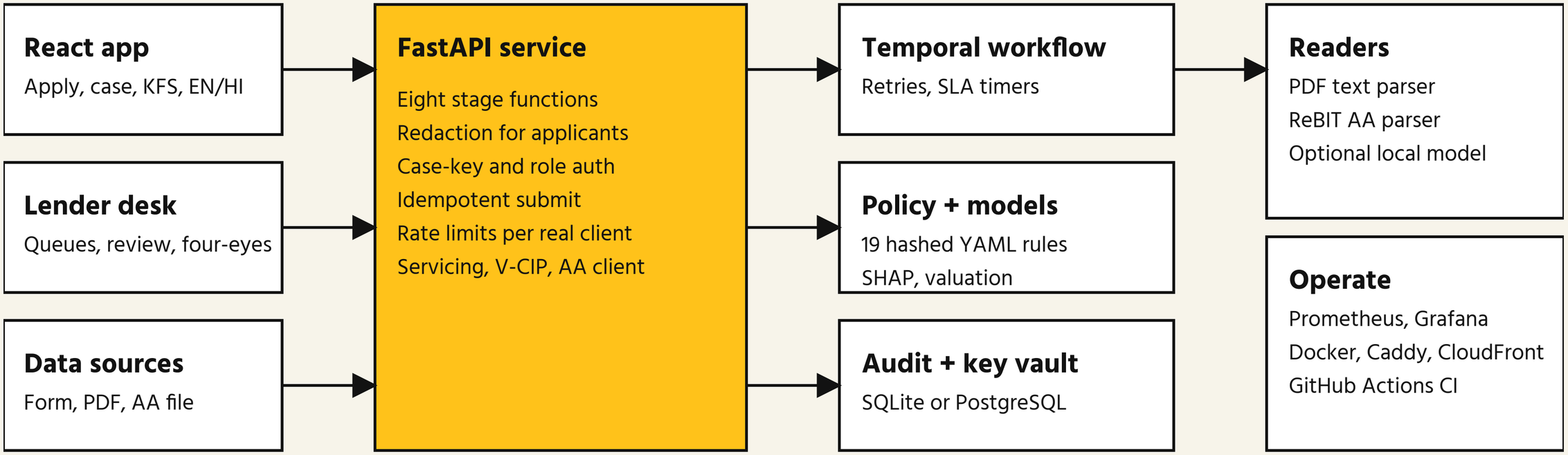
86 of 1000 decisions change.

Metric	Current	Proposed	Change
Auto-approved	53.8%	58.8%	↑ 5.0%
Referred to officers	37.5%	32.9%	↓ 4.6%
Declined	8.7%	8.3%	↓ 0.4%

Policy simulator: gain and cost of a rule change together

ARCHITECTURE

A stateless engine, a durable workflow, and an audit log as the source of truth.



SECURITY AND SCALE

Built as if the data were real, and measured, not promised.

Encrypted

AES-256-GCM, one key per case

Private by link

Case key, or staff login; else "not found"

Tamper-evident

Hash-chained audit, replayable

Rights

Erase, access, 72-hour breach register

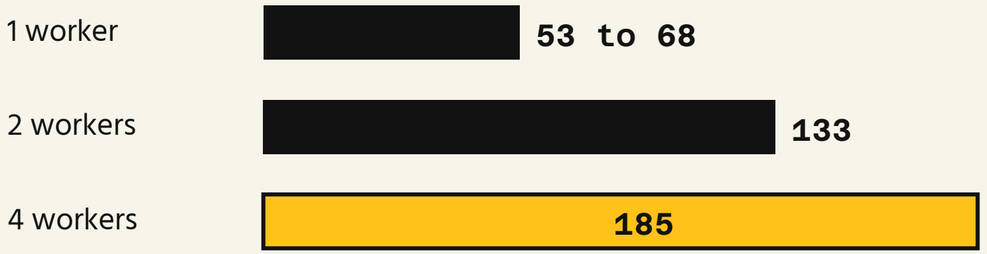
Hardened input

Strict schemas, XML entities refused, size caps

Clean scans

bandit, pip-audit, pnpm audit in CI

Decisions per second



Laptop. A 120 s run: 13,718 decisions, zero errors.

Not claimed: a million decisions a day. Peaks, bureau latency and the database would decide it.

HONESTY

What is real, what is simulated, and what is not built.

REAL

- APR and the Annex I statement, EN and HI
- Four sanctions lists, 31,347 entries
- Servicing and V-CIP rules
- AA signing and encryption
- Hash-chained audit, encryption
- DPDP rights, breach register

SIMULATED

- Bureau, KYC, land records
- Every applicant and statement
- The Aggregator: a mock
- V-CIP video: hashes only
- Property prices, default outcomes

NOT BUILT

- Live AA and FIU (needs RBI, Sahamati)
- MHA list, PEP data
- CKYCR upload (needs CERSAI)
- V-CIP recording, liveness model
- India-region hosting

No Indian loan-level data was available, so policy cut-offs are illustrative and the Hindi text awaits a legal translator. For a lender, `griha.byod --csv book.csv` retrains the same model on its own loan book. Sources for every figure: the Data sources page and README.

OPEN IT AND JUDGE FOR YOURSELF

Run a case. Open the reasons. Replay the decision.

LIVE DEMO
d3l1fg79dbflis.cloudfront.net

CODE, TESTS, DOCS
github.com/anshul23102/griha

A one-minute tour

1. Run a live case on the home page.
2. Open its Key Facts Statement; switch to Hindi.
3. Apply with the flaky-bureau test; watch the retry.
4. In Governance, verify the chain and replay a decision.

TeamBreezy
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A prototype on synthetic data, not a lending product or legal advice.